

Multiple integrals – 3

Volumes, sections and asymptotics in the dimension

13 сентября 2026 года

Theoretical minimum.

Theorem 1 (Cavalieri). $\text{vol}(A) = \int_{\mathbb{R}} \text{vol}_{n-1}(A \cap \{x_n = t\}) dt$: a volume is the integral of the areas of the sections. Together with the recursion $V_n = V_{n-1} \int_{-1}^1 (1 - t^2)^{\frac{n-1}{2}} dt$ this gives $V_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}$ for the unit ball.

Theorem 2 (what happens as n grows). $V_n \rightarrow 0$ faster than any geometric progression, the ratio of the volumes of the ball and of the circumscribed cube tends to zero, and the volume of the unit ball concentrates near the equator: for fixed $\varepsilon > 0$ the slab $|x_1| \leq \frac{\varepsilon}{\sqrt{n}}$ already contains almost all of it. In high dimension “almost all of the volume” always lives where the measure of a section is maximal (Convex – 9).

Fact 1 (volumes as probabilities). The volume of a subset of the cube $[0, 1]^n$ is the probability that n independent uniform points land in it, so questions about order statistics, about the maximum and the minimum of coordinates and about their spacings are volume computations — and conversely (Probability – 2).

Fact 2 (what is measured by a determinant). The volume of the parallelepiped spanned by $v_1, \dots, v_k$ in $\mathbb{R}^n$ equals $\sqrt{\det G}$, where G is the Gram matrix (Linear algebra – 5); the volume of the simplex on the same vectors is $\frac{1}{k!}$ of it. The inequalities about volumes — Brunn–Minkowski, John, the isoperimetric inequality — are in the series Convex – 4...10; here volumes are computed, not estimated.

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1. Prove Cavalieri’s principle, derive the recursion for the volume of the unit ball in $\mathbb{R}^n$ and the closed formula, and find the dimension in which V_n is maximal.

2. (МФТИ) Find the volume of the set of points $(x_1, \dots, x_n)$ of the cube $[0, 1]^n$ all of whose coordinates differ pairwise by more than d , where $d > 0$.
3. Prove that the volume of the unit ball in $\mathbb{R}^n$ concentrates near any equatorial section: for every $\varepsilon > 0$ the part of the ball with $|x_1| \geq \frac{\varepsilon}{\sqrt{n}}$ has relative volume tending to zero, and compute the precise Gaussian profile of the sections.
4. Compute the volume of the set $\{x \in [0, 1]^n : x_1 + \dots + x_n \leq t\}$ for every t , and deduce the probability that n independent uniform points on $[0, 1]$ have sum at most t ; in particular find $\lim_{n \rightarrow \infty} \text{vol}\{x \in [0, 1]^n : x_1 + \dots + x_n \leq \frac{n}{2}\}$.
5. Prove that the volume of the parallelepiped spanned by $v_1, \dots, v_k$ equals $\sqrt{\det G}$ and that the k -dimensional volume of the orthogonal projection of a body onto a subspace never exceeds the volume of the body divided by the distance it is spread over; then compute the maximal area of a plane projection of the unit cube in $\mathbb{R}^3$.